

Adham Ashraf Saeed

Assignment 6 (SQL)

Load the provided dataset into your database

- Access the following Database then Find:

Database

Questions

1. Total revenue by customer showing customer name and total revenue

```
CREATE VIEW Revenue_by_Customer
AS
(SELECT    C.name,
           SUM(O.amount) AS Total_Revenue
FROM      customers AS C
           INNER JOIN
           orders AS O
           ON O.customer_id = C.customer_id
GROUP BY C.name);
```

2. Customers with at least 3 orders showing customer name and order count

```
CREATE VIEW Customer_Orders
AS
(SELECT    C.name,
           COUNT(DISTINCT O.order_id) AS Total_Orders
FROM      customers AS C
           INNER JOIN
           orders AS O
           ON O.customer_id = C.customer_id
GROUP BY C.name
HAVING    COUNT(DISTINCT O.order_id) >= 3);
```


Questions

3. Average order value per customer showing customer name and average amount

```
CREATE VIEW AVG_Order_per_Customer
AS
(SELECT    C.name,
           AVG(O.amount) AS Total_Revenue
FROM      customers AS C
           INNER JOIN
           orders AS O
           ON O.customer_id = C.customer_id
GROUP BY  C.name);
```

4. Total revenue by product showing product name and total revenue

```
CREATE VIEW Revenue_by_Product
AS
(SELECT    P.product_name,
           SUM(O.amount) AS Total_Revenue
FROM      products AS P
           INNER JOIN
           orders AS O
           ON O.product_id = P.product_id
GROUP BY  P.product_name);
```


Questions

5. Total quantity sold by product showing product name and total quantity sold

```
CREATE VIEW Quantity_by_Product
AS
(SELECT    P.product_name,
           SUM(O.quantity) AS Total_Quantity
FROM      products AS P
           INNER JOIN
           orders AS O
           ON O.product_id = P.product_id
GROUP BY P.product_name);
```

6. Number of orders per day showing order date and order count

```
CREATE VIEW Orders_per_day
AS
(SELECT    order_date,
           COUNT(DISTINCT order_id) AS Total_Orders
FROM      orders
GROUP BY order_date);
```


Questions

7. Total sales per day showing order date and daily sales amount

```
CREATE VIEW Sales_per_day
AS
(SELECT    order_date,
           SUM(amount) AS Total_Orders
FROM      orders
GROUP BY  order_date);
```

8. Total payments by payment method showing payment method and total collected

```
CREATE VIEW payments_by_methods
AS
(SELECT    payment_method,
           SUM(payment_amount) AS TotalPayments
FROM      payments
GROUP BY  payment_method);
```


Questions

9. 9. Total payments per day showing payment date and total payment amount

```
CREATE VIEW payments_per_day
AS
(SELECT    payment_date,
           SUM(payment_amount) AS TotalPayments
FROM      payments
GROUP BY  payment_date);
```

10. Shipment summary by carrier showing carrier name, shipment count, first shipping date, and last shipping date

```
CREATE VIEW Shipment_summary
AS
(SELECT    carrier,
           COUNT(DISTINCT shipment_id) AS TotalShipment,
           MIN(shipping_date) AS First_shipping_date,
           MAX(shipping_date) AS Last_shipping_date
FROM      shipments
GROUP BY  carrier);
```


Bounce Problems

1. Orders with shipping status: order ID, customer name, order date, shipped_status (shipped / not_shipped)

```
CREATE VIEW Orders_with_ShippingSt
AS
(SELECT O.order_id,
       C.name,
       O.order_date,
       CASE
         WHEN S.shipment_id IS NOT NULL THEN 'Shipped'
         ELSE 'Not Shipped'
       END AS Shipped_Status
FROM   orders AS O
       INNER JOIN
       customers AS C
       ON C.customer_id = O.customer_id
       LEFT JOIN
       shipments AS S
       ON S.order_id = O.order_id);
```

2. Product price rank band: product name, price, price_band based on price (budget/standard/premium)

```
CREATE VIEW Product_price_band
AS
(SELECT product_id,
       product_name,
       price,
       CASE
         WHEN price <= 100 THEN 'Budget'
         WHEN price <= 500 THEN 'Standard'
         ELSE 'Premium'
       END AS Price_Band
FROM   products);
```